

Zenghuang Fu

MSc Student, Institute of Automation, Chinese Academy of Sciences

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Summary

- Master's student at the Institute of Automation, Chinese Academy of Sciences, advised by Weiliang Meng and Xiaopeng Zhang. My research centres on multimodal large models and agentic reinforcement learning, with a focus on long-horizon planning and memory; I also work on video understanding, MLLM reasoning, object detection and multi-object tracking. I am actively looking for PhD positions.

Education

MSc	Institute of Automation, Chinese Academy of Sciences , Pattern Recognition and Intelligent Systems	Beijing, China Sept 2024 – June 2027
	• Advised by Weiliang Meng and Xiaopeng Zhang.	

Publications

Robust detection in complex construction sites: HiPA-DETR with weather-aware and cross-domain generalization	May 2026
Zenghuang Fu, Jinming Yang, Xiaofeng Han, Zhe Feng, Muyang Zhang, Changwei Wang, Weiliang Meng, Jiguang Zhang, Xiaopeng Zhang (The Visual Computer)	
Cognitive Scaffold: From Fluid Context to Crystallized Memory for Long-Horizon Deep-Research Agents	July 2026
Qiuyuan Ai, Zenghuang Fu, Zhaoyang Li, Ping Jiang, Haoyu Wu, Jie Song, Guannan He (ACL (Main))	
WM-DETR: Dual-Branch Wavelet-Mamba and Sparse Attention for Robust Underwater Object Detection	June 2026
Xiaofeng Han, Shunpeng Chen, Zenghuang Fu, Longzhao Huang, Shengpeng Xu, Jinming Yang, Yixian Kong, Changwei Wang, Weiliang Meng, Xiaopeng Zhang (Expert Systems with Applications)	
Explicit temporal-semantic modeling for dense video captioning via context-aware cross-modal interaction	Mar 2026
Mingda Jia, Weiliang Meng, Zenghuang Fu, Yiheng Li, Qi Zeng, Yifan Zhang, Ju Xin, Rongtao Xu, Jiguang Zhang, Xiaopeng Zhang (AAAI)	
HMR-1: Hierarchical Massage Robot with Vision-Language-Model for Embodied Health-care	Mar 2026
Rongtao Xu, Mingming Yu, Xiaofeng Han, Yu Zhang, Kaiyi Hu, Zhe Feng, Zenghuang Fu, Changwei Wang, Weiliang Meng, Xiaopeng Zhang arxiv.org/abs/2603.08817 (arXiv preprint arXiv:2603.08817)	
DMTrack: Deformable State-Space Modeling for UAV Multi-Object Tracking with Kalman Fusion and Uncertainty-Aware Association	Oct 2025
Zenghuang Fu, Xiaofeng Han, Mingda Jia, Qi Zeng, Muyang Zhang, Changwei Wang, Weiliang Meng, Xiaopeng Zhang arxiv.org/abs/2510.17860 (arXiv preprint arXiv:2510.17860)	

Multimodal fusion and vision-language models: A survey for robot vision

Aug 2025

Xiaofeng Han, Shunpeng Chen, Zenghuang Fu, Zhe Feng, Lue Fan, Dong An, Changwei Wang,
Li Guo, Weiliang Meng, Xiaopeng Zhang, Rongtao Xu, Shibiao Xu
(Information Fusion)

Projects

Awesome-Long-Horizon-Credit-Assignment-for-Agentic-RL

A curated reading list of papers, methods and resources on long-horizon credit assignment for agentic RL.

- <https://github.com/Zenghuang-Fu/Awesome-Long-Horizon-Credit-Assignment-for-Agentic-RL>

DMTrack

Reference implementation of the DMTrack UAV multi-object tracker.

- <https://github.com/Zenghuang-Fu/DMTrack>

Research Interests

Multimodal: Multimodal large models, MLLM reasoning, vision-language models

Agents: Agentic reinforcement learning, long-horizon credit assignment, memory and planning

Perception: Video understanding, object detection, multi-object tracking

Technical Skills

Programming: Python, PyTorch, C++

Language Skills

Chinese: Native speaker

English: Professional working proficiency